

ECE 435 Homework #2 Solutions

Fall 2025

1. (a) $4e^{-j28z} \hat{y} \text{ V/m}$

(b) $-9je^{-j23z} \hat{y} \text{ V/m}$

(c) $7e^{-j23z} \hat{y} \text{ mA/m}$

(d) $4.6e^{-48\pi z} j(\frac{11}{4} - 139z) \hat{y} \text{ V/m}$

2. (a) $15 \cos(\omega t - 23z) \hat{x} \text{ V/m}$

(b) $32e^{-52z} \cos(\omega t - 135z) \hat{y} \text{ V/m}$

(c) $2.83 \cos(\omega t - 245z + 1.4) \hat{x} + 3.16 \cos(\omega t - 245z - 1.25) \hat{y} \text{ A/m}$

(d) $-3 \cos(\omega t) \text{ C/m}^3$

3. (a) $-\frac{4320 + j1664}{\omega \mu} e^{-(52 + j135)z} \hat{x} \text{ A/m}$

(b) $\frac{1}{\omega \epsilon} [(244.6 - j732.7) \hat{x} - (490 - j490) \hat{y}] e^{-245z} \text{ V/m}$

4. $\bar{S} = 0 \text{ W/m}^2$

5. Follow the steps used to derive the Helmholtz equation for the electric field in the notes. This is a derivation so no steps should be left out.